ISRO Computer Science Mock Test 1 - 2025

Instructions:
- **Total Questions:** 95 (80 Technical + 15 Aptitude)
- **Time:** 120 minutes
- **Marking:** +1 for correct, -1/3 for wrong, 0 for unattempted
- **All questions are compulsory**

PART A - TECHNICAL QUESTIONS (80 Questions)

Data Structures and Algorithms (Q1-20)

1. The worst-case time complexity of finding the kth smallest element in an unsorted array using quickselect is:
(a) O(n) (b) O(n log n) (c) $O(n^2)$ (d) O(k log n)

2. In a min-heap with n elements, the time complexity to delete an arbitrary element is:
(a) O(1) (b) O(log n) (c) O(n) (d) O(n log n)

3. Which traversal of a binary tree gives nodes in non-decreasing order for a Binary Search Tree?
(a) Preorder (b) Inorder (c) Postorder (d) Level order

4. The minimum number of edges that must be removed to make a connected graph with n vertices acyclic is:
(a) n-1 (b) n (c) e-n+1 (where e is edges) (d) e-n

5. In dynamic programming, which principle is primarily used?
(a) Greedy choice (b) Optimal substructure (c) Divide and conquer (d) Backtracking

6. The space complexity of merge sort is:
(a) O(1) (b) O(log n) (c) O(n) (d) O(n log n)

7. Which data structure is best for implementing undo functionality in text editors?
(a) Queue (b) Stack (c) Deque (d) Priority Queue

8. The maximum number of nodes in a complete binary tree of height h is:
(a) 2^h - 1 (b) 2^h (c) 2^(h+1) - 1 (d) 2^(h-1)

9. Floyd-Warshall algorithm finds:
(a) Single source shortest path (b) Minimum spanning tree (c) All pairs shortest path (d) Maximum flow

10. The amortized time complexity of union operation in Union-Find with path compression is:
(a) O(1) (b) O(log n) (c) $O(\alpha(n))$ (d) O(n)

11. Which sorting algorithm performs best on nearly sorted arrays?
(a) Quick Sort (b) Heap Sort (c) Insertion Sort (d) Selection Sort

12. In a hash table with chaining, if the load factor is λ, the expected time for unsuccessful search is:
(a) $O(1)$ (b) $O(\lambda)$ (c) $O(1 + \lambda)$ (d) $O(\log \lambda)$

13. The maximum height of an AVL tree with n nodes is approximately:
(a) log n (b) 1.44 log n (c) 2 log n (d) n

14. Which graph traversal can be used to detect cycles in an undirected graph?
(a) BFS only (b) DFS only (c) Both BFS and DFS (d) Neither

15. The time complexity of building a heap from an unsorted array is:
(a) $O(n \log n)$ (b) $O(n)$ (c) $O(\log n)$ (d) $O(n^2)$

16. In Dijkstra's algorithm, which data structure provides the best performance?
(a) Array (b) Linked List (c) Binary Heap (d) Fibonacci Heap

17. The worst-case time complexity of search in a skip list is:
(a) $O(1)$ (b) $O(\log n)$ (c) $O(n)$ (d) $O(\sqrt{n})$

18. Which tree traversal uses a stack data structure in its iterative implementation?
(a) Level order (b) Inorder (c) Preorder (d) Both (b) and (c)

19. The minimum number of colors needed to color any planar graph is at most:
(a) 3 (b) 4 (c) 5 (d) 6

20. In a B-tree of order m, the maximum number of keys in a non-root node is:
(a) m (b) m-1 (c) m+1 (d) 2m-1

Programming Languages and Software Engineering (Q21-30)

21. In object-oriented programming, which concept allows a single interface to represent different underlying forms?
(a) Inheritance (b) Encapsulation (c) Polymorphism (d) Abstraction

22. Which design pattern ensures a class has only one instance?
(a) Factory (b) Observer (c) Singleton (d) Strategy

23. In the software development lifecycle, which phase comes immediately after requirements analysis?
(a) Implementation (b) Testing (c) Design (d) Maintenance

24. What is the main advantage of using interfaces over abstract classes in Java?
(a) Better performance (b) Multiple inheritance (c) Easier syntax (d) Automatic memory management

25. Which testing technique focuses on the internal structure of the program?
(a) Black box testing (b) White box testing (c) Gray box testing (d) Integration testing

26. In version control systems, what is a "merge conflict"?
(a) Hardware failure (b) Network error (c) Simultaneous changes to same code (d) Memory overflow

27. What does the acronym SOLID represent in software engineering?
(a) System design principles (b) Testing methodologies (c) Database rules (d) Network protocols

28. Which software development methodology emphasizes iterative development?
(a) Waterfall (b) Spiral (c) V-Model (d) Agile

29. In functional programming, what is a "pure function"?
(a) Function with no parameters (b) Function with no side effects (c) Function returning void (d) Function using only primitives

30. What is the purpose of a compiler's lexical analyzer?
(a) Syntax checking (b) Code optimization (c) Token generation (d) Error reporting

Computer Networks (Q31-45)

31. In the TCP/IP model, which layer handles end-to-end communication?
(a) Network (b) Transport (c) Application (d) Link

32. What is the maximum window size in TCP with a 16-bit window field?
(a) 65535 bytes (b) 65536 bytes (c) 32768 bytes (d) 1 MB

33. Which routing protocol uses the Bellman-Ford algorithm?
(a) OSPF (b) RIP (c) BGP (d) EIGRP

34. In Ethernet, what does CSMA/CD stand for?
(a) Carrier Sense Multiple Access/Collision Detection (b) Circuit Switch Multiple Access/Code Division (c) Channel State Multiple Access/Collision Defragment (d) Carrier Signal Multiple Access/Channel Division

35. What is the primary function of ARP (Address Resolution Protocol)?
(a) IP to MAC mapping (b) Domain name resolution (c) Route discovery (d) Error detection

36. In a subnet with mask 255.255.255.192, how many host addresses are available?
(a) 62 (b) 64 (c) 126 (d) 128

37. Which protocol operates at the application layer and transfers files?
(a) HTTP (b) FTP (c) SMTP (d) All of the above

38. What is the purpose of NAT (Network Address Translation)?

(a) Speed up internet (b) Provide security (c) Map private to public IPs (d) Reduce network traffic

39. In wireless networks, what does WPA stand for?
(a) Wireless Personal Access (b) Wireless Protected Access (c) Wireless Private Access (d) Wireless Protocol Access

40. The sliding window protocol is used for:
(a) Error detection (b) Flow control (c) Routing (d) Encryption

41. What is the default port number for HTTPS?
(a) 80 (b) 443 (c) 8080 (d) 21

42. In IPv6, what is the address length?
(a) 32 bits (b) 64 bits (c) 128 bits (d) 256 bits

43. Which layer of the OSI model handles encryption?
(a) Session (b) Presentation (c) Application (d) Transport

44. What is the maximum data rate of 802.11n wireless standard?
(a) 54 Mbps (b) 150 Mbps (c) 300 Mbps (d) 600 Mbps

45. In DNS, what does the MX record represent?
(a) Mail exchange (b) Maximum transfer (c) Memory extension (d) Media exchange

Operating Systems (Q46-60)

46. Which scheduling algorithm can cause starvation?
(a) FCFS (b) Round Robin (c) Priority Scheduling (d) SJF

47. In paging, what is the purpose of the Translation Lookaside Buffer (TLB)?
(a) Store pages (b) Cache page table entries (c) Manage virtual memory (d) Handle page faults

48. Which replacement algorithm suffers from Belady's anomaly?
(a) LRU (b) FIFO (c) Optimal (d) Clock

49. What is a race condition in operating systems?
(a) CPU scheduling conflict (b) Memory allocation error (c) Unsynchronized access to shared resource (d) Network timeout

50. In the producer-consumer problem, what synchronization primitive is most appropriate?
(a) Mutex (b) Semaphore (c) Monitor (d) Spinlock

51. Which memory allocation technique suffers from external fragmentation?
(a) Paging (b) Segmentation (c) Virtual memory (d) Cache memory

52. What is the purpose of the working set model?
(a) CPU scheduling (b) Memory management (c) File systems (d) I/O management

53. In Unix/Linux, what does the chmod 755 command do?
(a) Changes owner (b) Sets permissions (c) Creates directory (d) Modifies path

54. Which system call is used to create a new process?
(a) exec() (b) fork() (c) wait() (d) exit()

55. What is thrashing in virtual memory systems?
(a) High CPU utilization (b) Memory corruption (c) Excessive page faulting (d) Disk failure

56. In deadlock prevention, which condition is addressed by resource preemption?
(a) Mutual exclusion (b) Hold and wait (c) No preemption (d) Circular wait

57. What is the difference between hard and soft real-time systems?
(a) Hardware requirements (b) Deadline strictness (c) Memory usage (d) CPU speed

58. Which file system is commonly used in Linux?
(a) NTFS (b) FAT32 (c) ext4 (d) HPFS

59. What is the purpose of system calls?
(a) Hardware communication (b) User-kernel interface (c) Process communication (d) Memory management

60. In symmetric multiprocessing (SMP), what is shared among processors?
(a) Memory and I/O (b) Cache only (c) Nothing (d) Registers only

Database Management Systems (Q61-70)

61. In RDBMS, what does ACID stand for?
(a) Atomicity, Consistency, Isolation, Durability (b) Access, Control, Integration, Distribution (c) Authentication, Confidentiality, Integrity, Data (d) Analysis, Compilation, Implementation, Design

62. Which normal form eliminates transitive dependencies?
(a) 1NF (b) 2NF (c) 3NF (d) BCNF

63. In SQL, which join returns all records when there is a match in either table?
(a) INNER JOIN (b) LEFT JOIN (c) RIGHT JOIN (d) FULL OUTER JOIN

64. What is the purpose of indexing in databases?
(a) Data compression (b) Query optimization (c) Security enhancement (d) Backup creation

65. Which concurrency control technique uses timestamps?
(a) Locking (b) Timestamp ordering (c) Validation (d) Multiversion

66. In ER modeling, what represents the relationship between entities?
(a) Rectangle (b) Ellipse (c) Diamond (d) Circle

67. What is a deadlock in database transactions?
(a) System crash (b) Circular dependency (c) Memory overflow (d) Network failure

68. Which SQL command is used to modify table structure?
(a) UPDATE (b) ALTER (c) MODIFY (d) CHANGE

69. What is the difference between DELETE and TRUNCATE?
(a) No difference (b) DELETE is faster (c) TRUNCATE removes all rows (d) DELETE uses less memory

70. In NoSQL databases, what does CAP theorem state?
(a) Performance guarantees (b) Security requirements (c) Consistency-Availability-Partition tolerance trade-offs (d) Storage limitations

Computer Architecture and Digital Logic (Q71-80)

71. In a 4-bit binary number, what is the range of values?
(a) 0-15 (b) 0-16 (c) 1-16 (d) -8 to +7

72. What is the function of ALU in a CPU?
(a) Memory management (b) Arithmetic and logic operations (c) I/O control (d) Instruction decoding

73. In cache memory, what does "cache hit" mean?
(a) Memory error (b) Data found in cache (c) Cache overflow (d) Cache miss

74. Which logic gate produces output 0 only when both inputs are 1?
(a) AND (b) OR (c) NAND (d) NOR

75. What is the purpose of virtual memory?
(a) Increase CPU speed (b) Extend physical memory (c) Improve graphics (d) Enhance security

76. In instruction pipelining, what is a hazard?
(a) Hardware failure (b) Pipeline stall condition (c) Memory error (d) Power failure

77. What does RISC stand for in computer architecture?
(a) Reduced Instruction Set Computer (b) Rapid Instruction Set Computer (c) Real Instruction Set Computer (d) Random Instruction Set Computer

78. In memory hierarchy, which has the fastest access time?
(a) Main memory (b) Cache (c) Registers (d) Secondary storage

79. What is the binary representation of decimal 13?
(a) 1011 (b) 1101 (c) 1110 (d) 1010

80. In floating-point representation, what does the mantissa represent?
(a) Sign bit (b) Exponent (c) Fractional part (d) Whole number

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PART B - APTITUDE/ABILITY TEST (15 Questions)

81. If COMPUTER is coded as RFUVQNFQ, then SCIENCE is coded as:
(a) FPJFODF (b) VDJFODF (c) TDJEFODF (d) VDJODEF

82. What comes next in the sequence: 2, 6, 12, 20, 30, ?
(a) 40 (b) 42 (c) 45 (d) 48

83. In a group of 50 people, 30 like tea, 25 like coffee, and 10 like both. How many like neither?
(a) 5 (b) 10 (c) 15 (d) 20

84. A train travels 300 km in 5 hours. What is its speed in m/s?
(a) 16.67 (b) 60 (c) 20 (d) 25

85. If $\log_2(x) = 3$, then x equals:
(a) 6 (b) 8 (c) 9 (d) 12

86. Find the missing number: 5, 11, 23, 47, ?
(a) 94 (b) 95 (c) 96 (d) 97

87. In how many ways can the letters of ALGORITHM be arranged?
(a) 9! (b) 9!/2! (c) 8! (d) 9!/2! × 2!

88. If 3x + 7 = 22, then x equals:
(a) 5 (b) 6 (c) 7 (d) 8

89. What is 25% of 80?
(a) 15 (b) 20 (c) 25 (d) 30

90. A rectangle has length 12 cm and breadth 8 cm. Its area is:
(a) 96 cm^2 (b) 40 cm^2 (c) 20 cm^2 (d) 48 cm^2

91. Complete the series: ACE, FHJ, KMO, ?
(a) PRT (b) PSU (c) PRS (d) PQS

92. If a clock shows 3:15, what is the angle between hour and minute hands?
(a) 7.5° (b) 15° (c) 22.5° (d) 30°

93. The sum of first 10 natural numbers is:
(a) 45 (b) 50 (c) 55 (d) 60

94. If the ratio of boys to girls in a class is 3:2 and there are 15 boys, how many girls are there?
(a) 8 (b) 10 (c) 12 (d) 15

95. What is the next number in the pattern: 1, 4, 9, 16, 25, ?
(a) 30 (b) 35 (c) 36 (d) 49

ANSWER KEY

Technical Questions (1-80):
1-c, 2-c, 3-b, 4-c, 5-b, 6-c, 7-b, 8-c, 9-c, 10-c, 11-c, 12-c, 13-b, 14-c, 15-b, 16-d, 17-c, 18-d, 19-b, 20-b, 21-c, 22-c, 23-c, 24-b, 25-b, 26-c, 27-a, 28-d, 29-b, 30-c, 31-b, 32-a, 33-b, 34-a, 35-a, 36-a, 37-d, 38-c, 39-b, 40-b, 41-b, 42-c, 43-b, 44-d, 45-a, 46-c, 47-b, 48-b, 49-c, 50-b, 51-b, 52-b, 53-b, 54-b, 55-c, 56-c, 57-b, 58-c, 59-b, 60-a, 61-a, 62-c, 63-d, 64-b, 65-b, 66-c, 67-b, 68-b, 69-c, 70-c, 71-a, 72-b, 73-b, 74-c, 75-b, 76-b, 77-a, 78-c, 79-b, 80-c

Aptitude Questions (81-95):
81-b, 82-b, 83-a, 84-a, 85-b, 86-b, 87-a, 88-a, 89-b, 90-a, 91-a, 92-a, 93-c, 94-b, 95-c